

Programming Languages for Blockchain are Here to Stay

Blockchain technologies are currently being used in many spheres of life apart from the finance sector — in healthcare, record-keeping, logistics, etc. Blockchains are written in many programming languages. This article takes a look at some of them.



in blockchain technologies, the industry looks set to gain further momentum.

There are dedicated programming languages for blockchain related technologies. The abundance of programming languages has always amazed me. Developers are willing to create a new programming language rather than modify an existing one, in order to suit a new industry, and that has led to many exciting new options. Blockchain programming languages like Solidity, Simplicity, Vyper, CX, Scilla, Rholang and Obsidian are some of them. Readers will notice that I have omitted the popular Bitcoin Script from this list since it is not Turing complete because it does not support loops. But the unavailability of loops is not an accident. It is done intentionally to avoid the occurrence of infinite loops.

I have tried to work with just one among the programming languages mentioned above, called Solidity, for which I was able to easily install the compiler in my system.

Solidity

Solidity is an object-oriented programming language which was developed for creating smart contracts, an essential part of blockchain technologies.



Figure 1: The logo of the Solidity programming language

The first time I heard about cryptocurrencies was in 2011 when I read an article about Bitcoin. To be frank, I was sceptical about the future of a currency whose value was not tied to any real-world entity, like gold. In March 2013, it was reported that some bugs were found in version 0.8 of the Bitcoin software. I believe that was the first time Bitcoin was widely covered by the press in India. Many experts predicted the end of Bitcoin and cryptocurrencies, with some comparing Bitcoins with the legendary tulip mania—the prices for some bulbs of the tulip (a type of flower) reached extraordinarily high

levels in the then Dutch Republic but soon collapsed in February 1637. This was also the first recorded instance of a speculative bubble. But the fate of Bitcoin was different. Time and again we have seen that predicting the future of a technology is a risky business.

Getting back to cryptocurrencies, today, after more than seven years (as of May 2020) the cryptocurrency market is a whopping US\$ 1 billion industry. Bitcoin now has many competitors like Ether, Ripple, etc. Besides, the blockchain has found applications in diverse areas like smart contracts, healthcare, personal identification, etc. With major multinational companies investing heavily

